

AddCoinToCountAndPlot

Create a function **AddCoinToCountAndPlot**, which you will use in your final project script. The function takes as input **x** and **y**, coordinate image, **cls**, its classification label indicating whether it was found to be a dime, nickel, or quarter. The function outputs **coinvalue**, the value also plots, in the current figure, a circle centered at **x** and **y** with radius and color unique for each coin type according to the table below. 1

Function ?

Save

Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```

1 function [coinvalue,x_plot,y_plot,col] = AddCoinToPlotAndCount(x,y,cls)
2 % initialize radians for defining x_plot and y_plot using cos and sin functions
3 rads = 0:2*pi/32:2*pi;
4 % initialize parameters for radius and color of circle for each type of coin
5 % initialize parameters for radius and color of circle for each type of coin
6 r = 22;
7 col = 'r';
8 % use if-elseif statement to define x_plot, y_plot, col
9 %   when cls is 1, we found a dime
10 if cls == 1
11     coinvalue = 10;
12     r = 22;
13     col = 'r';
14 %   when cls is 2, we found a nickel
15 elseif cls == 2
16     coinvalue = 5;
17     r = 30;
18     col = 'g';
19 %   when cls is 3, we found a quarter
20 elseif cls == 3
21     coinvalue = 25;
22     r = 40;
23     col = 'm';
24 end
25 x_plot = x+r*cos(rads);
26 y_plot = y+r*sin(rads);
27 plot(x_plot,y_plot,col);
28 end
29

```

Code to call your function ?

Reset

```

1 x=1.2;
2 y=4.1;
3 cls = 3;
4 [coinvalue,x_plot,y_plot,col] = AddCoinToPlotAndCount(x,y,cls)
5 figure
6 plot(x_plot,y_plot,[col,'o-'])

```

Run Function

?

Assessment: All Tests Passed

Submit ?

Is coinvalue correct?

Is x_plot correct for random centroid?

Is y_plot correct for random centroid?

